*SOLAR TII, Ltd*
*58, Nezavisimosti ave*
*Minsk 220005*
*Republic of Belarus*
*Phone: +375(17) 281-48-99*
*Fax: +375(17) 281-47-99*
*E-mail: sales@solartii.com*
*Web: www.solartii.com*

# MONOCHROMATOR-SPECTROGRAPH IMAGING
# MS 3501i s/n 011076
## Release certificate

| | |
|---|---|
| 1. List of components | 2 |
| 2. Final test sheet | 3 |

## 1. LIST OF COMPONENTS

| | Quantity (pce) |
|---|---|
| 1. Housing | |
| - with imaging aluminum optics Al+$MgF_2$ | 1 |
| - with imaging golden optics | |
| 2. Grating positioner | |
| - single (with manual change of gratings) | 1 |
| 3. Gratings | 1 |
| 4. Slits | |
| Combined | |
| - entrance | |
| - exit | |
| Manual | |
| - entrance | 1 |
| - exit | |
| 5. Filter wheel FWS8 (with 5 filters) with driver | |
| 6. Automated exit port selection mirror | |
| 7. Fast shutter | |
| 8. Fast shutter with imaging condenser | |
| 9. Condenser – F/#-matcher | |
| 10. Diaphragm with Holder to adjust Entrance Slit Height | |
| 11. Fiber adapter | |
| 12. Quartz optical fiber | |
| 13. Adapter for exit slit | |
| 14. Adapter for CCD - detector | |
| 15. Adapter for PMT - detector | |
| 16. Special condenser for C9940 attachment | |
| 17. Hand Controller | |
| 18. Detector LD 1024 USB (UV) 14 bit | |
| 19. Detector LD 2048 USB (UV) 12 bit | |
| 20. Detector LD 3648 USB (UV) 14 bit | |
| 21. Detector LD 512 USB (IR) 16 bit | |
| 22. Detector IGA2.2-010-TE2-H | |
| 23. Detector PMT R928 (UV) | |
| 24. Leg for CCD - detector | |
| 25. ADC board 16 bit | |
| 26. CD with **Software for MS 3504i** | 1 |
| 27. CD with Software **PSI-Line** | |
| 28. Mains Power Cable | 1 |
| 29. RS232C Interface Cable | 1 |

## 2. FINAL TEST SHEET

| | |
|---|---|
| Groove Spacing, lines/mm | 1800 |
| Blaze Wavelength, nm | 500 |
| Type | ruled |
| Coating | Al+$MgF_2$ |
| Wavelength range (>40% eff.), nm | 335-878 |
| Wavelength range (usable), nm | 190-878, zero order |
| Reciprocal dispersion, m/mm (at blaze wavelength) | 1,518 |
| *Wavelength resolution, nm | 0,055 |
| Wavelength accuracy, nm | See table below |
| Wavelength repeatability, nm | ±0,01 |

* Taken by CCD with pixel size 14 µm, entrance slit width 15 µm, at blaze wavelength.

Grating 1800 L/mm, blaze 500 nm

| Wavelength tabular, nm | Wavelength experimental, nm | Wavelength error, nm |
|---|---|---|
| 312,567 | 312,564 | -0,003 |
| 365,015 | 365,016 | 0,001 |
| 404,656 | 404,656 | 0 |
| 435,833 | 435,851 | 0,018 |
| 491,607 | 491,674 | 0,067 |
| 546,073 | 546,093 | 0,02 |
| 579,066 | 579,047 | -0,019 |
| 625,134 | 625,114 | -0,02 |
| 730,03 | 730,046 | 0,016 |
| 809,312 | 809,313 | 0,001 |
| 871,666 | 871,665 | -0,001 |